Polynomial Illumination Correction for the Arsenic Colorimeter: Analysis, Criteria, Implementation and Protocol

Development Notes

2026-08-04

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1 Purpose and scope

The colorimeter reads a chemical test strip by comparing the mean brightness of a small *sample* patch against the mean brightness of a *white reference* patch elsewhere on the same card. That comparison is meaningful only if the two patches are lit equally. They are not. Measured on the captures described below, the illumination inside the imaged region falls by about 25 % from the top of the region to the bottom.[1]

This document covers the correction for that non-uniformity. The correction map is called a *flat field*[2], and the decision documented here is to represent it as a low-order polynomial surface, fitted separately for each LED color and each Bayer layer.

The document reports:

- the notation and the definition of every measure used (§2);
- what the correction is and where it sits in the pipeline (§3);
- whether the correction has to be a map at all, or whether two numbers would do (§4);
- the polynomial model and why it takes the form it does (§5);
- the criteria proposed for judging a fit adequate (§6);
- what was measured, on which captures (§8);
- the choices those measurements drove (§9);
- how the evaluation harness is built (§10);
- the image sampling protocol for the captures still needed (§11);
- how to run the programs and read their output (§12, §13);
- what this unblocks (§14).

Nothing described here has been wired into the instrument's capture or analysis path. The work so far is a desktop evaluation harness, `devdesk/flatfield.py`, plus the archive metadata needed to make its inputs self-describing.

2 Notation, vocabulary and measures

2.1 Terms

Bayer layer
The camera sensor has one color filter per photosite, in a repeating 2×2 tile: red, two greens, blue. Splitting the raw image on that tile gives four half-resolution images, called *layers* and labeled `R`, `Gr`, `Gb`, `B`. `Gr` is the green photosite that shares a row with red; `Gb` is the one that shares a row with blue. They are not interchangeable; see §8.7.2.

DN
Digital Number: the raw sensor count, before conversion to physical units. The sensor is 10-bit, so DN runs 0 to 1023.

Stream space
Pixel coordinates in the full camera frame, 2304×1296 for the IMX708 mode in use.

Layer space
Pixel coordinates within one Bayer layer. A rectangle $w \times h$ in stream space is $w/2 \times h/2$ in layer space. All analysis in this document is in layer space.

[1] Denominator: the mean over the whole imaged region. The figure is the difference between the mean of the top row of pixels and the mean of the bottom row, divided by that overall mean.

[2] **Flat field.** A map of how brightly the illumination lands on each pixel when the imaged object is uniform. Dividing a picture by its flat field removes the lighting pattern and leaves only the object. The name is from astronomy, where the calibration image is taken of a featureless ("flat") screen.

`bound`
: The rectangular region of interest that the capture path crops and stores. Everything outside it is discarded at capture time. Currently 196×660 in stream space, which is 98×330 per layer.

Crop
: The stored `bound` region of one Bayer layer, as an array of 330 rows by 98 columns. "The crop" always means this whole array.

Patch
: One of the small rectangles the analysis measures: `sample`, `white`, `chip1`, `chip2`. Each is 68×68 in stream space, so 34×34 in layer space, which is 1156 pixels. A patch is a real region of interest at a real location on the card. Patches are addressed in code by *role* (`ROLE_SAMPLE`, `ROLE_WHITE`, ...), never by name, because names are operator-editable.

Window
: One tile of a regular 34×34 grid laid over the crop, used only as a measuring grid for the block-residual statistic (§2.3.1). Windows are the same size as patches but are placed by the grid, not by where an ROI is. The 330×98 crop holds $9 \times 2 = 18$ whole windows.

Extent
: The set of pixels a polynomial was fitted over, written Ω. The extent may be the whole crop or some subset of it; the choices tried are listed in §7.

Dark frame
: A capture taken at the same exposure with the LEDs off. Subtracting it removes the sensor's electronic offset and thermal signal. All work here is on dark-subtracted data.

Suite
: One capture archive (`.h5`) holding, for every LED color, a ladder of exposures plus the matching dark frames.

In-sample
: A fit evaluated on the same capture it was fitted to. Optimistic by construction: the fit has already seen that capture's noise and texture.

Out-of-sample, or transfer
: A fit taken from one capture and applied to a different one. This is what the instrument does in service, because it applies a stored calibration to a new measurement. It is therefore the estimate that predicts field performance.

2.2 Symbols

Every quantity below is defined per Bayer layer, for one LED color at one exposure. There is no mixing of layers or colors anywhere in the analysis.

Symbol	Meaning
$p = (x, y)$	a pixel location in layer space; x is the column, y the row
$D(p)$	lit frame, DN
$D_0(p)$	dark frame at the same exposure, DN
$S(p) = D(p) - D_0(p)$	dark-subtracted signal, DN
$A(p)$	reflectance of the imaged target, dimensionless — the quantity wanted
$L(p)$	illumination delivered to p, DN per unit reflectance
Ω	the fit extent: a set of pixels
$\lvert\Omega\rvert$	the number of pixels in that set
$F(p)$	the fitted polynomial surface, DN
$G(p)$	the gain map, dimensionless
$S^c(p)$	corrected signal, DN
R_i	the set of pixels in patch i; $n_i = \lvert R_i\rvert$, and $n_i = 1156$
w_i, h_i	the width and height of patch i in layer pixels; both 34
$\bar{s}_i$	the corrected mean of patch i, DN
$\bar{g}_i$	the mean of the gain map over patch i, dimensionless
W_k	the k-th window of the measuring grid

The angle-bracket notation means an arithmetic mean over a set of pixels:

$$\langle X \rangle_\Omega \equiv \frac{1}{|\Omega|} \sum_{p \in \Omega} X(p). \tag{1}$$

The subscript names the set. $\langle X \rangle_\Omega$ is a scalar.

The model is a pixelwise one throughout. D, D_0, S, A, L, F, G and S^c are all arrays of the same shape as the crop, and every operation combining them — subtraction in Eq. 9, division in Eq. 11 — is elementwise, one pixel at a time. No matrix product appears anywhere in the correction. The only scalars are the means defined by Eq. 1 and the polynomial coefficients.

2.3 The three measures

Three numbers recur throughout the results. All three are relative quantities expressed as percentages, and the denominator of each is part of its definition.

2.3.1 Block residual — how flat the corrected crop is

Lay the 34×34 measuring grid over the corrected crop and take the mean of each window:

$$m_k = \langle S^c \rangle_{W_k}. \tag{2}$$

If the correction were perfect and the imaged card uniform, every m_k would equal the mean of the whole corrected crop. Express each window's departure from that as a percentage:

$$d_k = 100 \left(\frac{m_k - \langle S^c \rangle_{\text{crop}}}{\langle S^c \rangle_{\text{crop}}} \right). \tag{3}$$

The ratio $m_k / \langle S^c \rangle_{\text{crop}}$ equals 1 when the window matches the whole crop, so subtracting 1 converts it from a ratio into a deviation, and multiplying by 100 expresses that deviation as a percentage. The denominator is the mean of the whole corrected crop.

Two summaries of $\{d_k\}$ are reported: the root-mean-square over the 18 windows, printed as `blk rms`, and the largest $|d_k|$, printed as `blk max`.

Averaging over 1156 pixels reduces the shot-noise contribution to about 0.04 % of the window mean, so what survives in d_k is spatial structure, not pixel noise.

2.3.2 Criterion P — patch-ratio accuracy

$\bar{s}_i$ is the corrected mean of patch i, that is $\langle S^c \rangle_{R_i}$, a single number in DN. $\bar{s}_{\text{white}}$ is the same quantity for the patch in the white-reference role, and it is the denominator.

Image a card that is genuinely uniform. Every patch should then read the same as the white reference, so for each patch i the quantity

$$P_i \;=\; 100 \left| \frac{\bar{s}_i - \bar{s}_{\text{white}}}{\bar{s}_{\text{white}}} \right| \tag{4}$$

should be zero, and whatever it is instead is the flat field's error.

Two levels of aggregation follow, and they are easy to confuse:

1. Within one frame, take the worst patch: $P = \max_i P_i$, over the patches present.
2. Across the set of frames scored — one frame per (LED color × Bayer layer) — report the median of P as `P med` and the maximum of P as `P max`.

So `P med` $= 0.24\,\%$ means: in half the 39 frames scored, the worst-deviating patch deviated by less than $0.24\,\%$ of that frame's corrected white-patch mean, and in half of them by more. `P max` is the single worst patch in the single worst frame. Neither figure is a median over patches.

2.3.3 Criterion S — residual tilt inside a patch

Fit a plane to the corrected patch by least squares,

$$\hat{v}(x, y) \;=\; a_0 + a_x\, x + a_y\, y, \qquad (x, y) \in R_i, \tag{5}$$

where x runs $0 \ldots w_i - 1$ and y runs $0 \ldots h_i - 1$ in layer pixels within the patch. The constant a_0 carries the patch level; a_x and a_y are the slopes, in DN per layer pixel.

A plane over a rectangle attains its largest and smallest values at opposite corners, so the peak-to-peak range of $\hat{v}$ over the patch is

$$R_{\text{plane}} \;=\; |a_x|\,(w_i - 1) \;+\; |a_y|\,(h_i - 1), \tag{6}$$

in DN. The spans are $w_i - 1$ and $h_i - 1$, not w_i and h_i, because a row of w_i pixels spans $w_i - 1$ intervals between first and last pixel center. Using w_i would overstate R_{plane} by a factor $w_i/(w_i - 1)$, here 34/33.

Criterion S is that range as a percentage of the patch's own mean:

$$S_i \;=\; 100\, \frac{R_{\text{plane}}}{\bar{s}_i}. \tag{7}$$

The denominator is patch i's own corrected mean, not the white patch's.

S_i needs a companion figure, because a plane fitted to pure noise still has non-zero slopes. Take n pixels evenly spaced along a side of length ℓ, with independent noise of standard deviation σ_v on each. The least-squares slope has standard error $\sigma_v \sqrt{12/(n\,\ell^2)}$, so the slope multiplied by the side length has standard error $\sigma_v \sqrt{12/n}$. The two axes contribute one such term each, and Eq. 6 adds them, so the range expected from noise alone is

$$S_i^{\text{nz}} \;=\; 2\,\sigma_i\, \sqrt{12/n_i}, \tag{8}$$

where σ_i is the pixel standard deviation within patch i as a percentage of $\bar{s}_i$. For $n_i = 1156$ the factor $2\sqrt{12/n_i}$ is 0.204.

The reported column `S/nz` is the ratio S_i / S_i^{nz}. A value near 1 means the tilt is what noise alone would have produced, so it is a fitting artifact. A value well above 1 means the corrected patch really does carry a tilt.

2.4 Other percentages

Percentages that do not fit one of the three definitions above are spelled out where they appear. The recurring ones are the masked-pixel fraction (denominator: pixels in the fit extent) and brightness ratios between two cards (denominator: named at the point of use).

3 Where the correction sits

The pipeline for one measurement, per Bayer layer:

1. Capture the lit frame D and its dark frame D_0 at the same exposure.
2. Subtract, pixel by pixel:
$$S(p) \;=\; D(p) - D_0(p). \tag{9}$$
3. Mask pixels that are saturated or dead (§5.4).
4. Divide S by the gain map G — this document's subject.
5. Take the patch means $\bar{s}_{\text{sample}}$ and $\bar{s}_{\text{white}}$, and form their ratio.
6. Convert that ratio to concentration by the strip calibration.

Step 4 is the only step that knows anything about illumination geometry. Its sole purpose is to make step 5's ratio independent of *where on the card* the two patches sit.

3.1 The correction divides; it does not subtract

`colorimeter_methods.tex` specified subtracting a saved trend array. That is correct only to first order. Illumination acts multiplicatively on the dark-subtracted signal: a pixel that receives 10 % less light returns 10 % less signal, not a fixed number of DN less. Writing the signal as the product of what the target reflects and what the lamp delivers,

$$S(p) \;=\; A(p)\,L(p), \tag{10}$$

makes the point plain: L multiplies, so removing it requires division.

The size of the error from getting this wrong was measured. On the two 2026-08-03 captures, the gray card returned 0.27× the signal of the white card at the same exposure.[3] An additive trend calibrated on the white card, applied to the gray card, over-corrects by a factor of $1/0.27 \approx 3.7$.

The correction used here is therefore

$$S^c(p) \;=\; \frac{S(p)}{G(p)}, \qquad G(p) \;=\; \frac{F(p)}{\langle F\rangle_\Omega}. \tag{11}$$

Both operations are elementwise over pixels. F is the fitted polynomial surface in DN, $\langle F\rangle_\Omega$ is a single number, and G is dimensionless with mean exactly 1 over Ω. G is called a *gain map*.

Note which extent normalizes F. The denominator $\langle F\rangle_\Omega$ is the mean of the fitted surface over the extent it was fitted on, not over the whole crop. For a fit taken over the whole crop the two are the same. For a fit taken over a small window around one patch they are not, and using the wrong one rescales the result without any visible symptom. The code stores this number as `FlatField.norm_mean`.

4 Does the correction need to be a map?

The measurement of interest is a ratio of two patch means, and a patch mean is a single number. It is worth asking directly whether the whole gain *map* is needed, or whether two scalars would

[3] Ratio of dark-subtracted patch means, gray divided by white. Denominator: the white card's patch mean.

give the same answer. The question turns out to have a clean answer, and answering it also explains two results reported later.

4.1 The exact case

Suppose the fit is perfect, so that $G = L/\langle L\rangle_\Omega$ exactly. Substituting Eq. 10 into Eq. 11,

$$S^c(p) \;=\; \frac{A(p)\,L(p)}{L(p)/\langle L\rangle_\Omega} \;=\; A(p)\;\langle L\rangle_\Omega\,. \tag{12}$$

The illumination cancels pixel by pixel, leaving reflectance times a constant. Taking patch means and forming the ratio,

$$\frac{\bar{s}_{\text{sample}}}{\bar{s}_{\text{white}}} \;=\; \frac{\langle A\rangle_{R_{\text{sample}}}}{\langle A\rangle_{R_{\text{white}}}}, \tag{13}$$

which is exactly the reflectance ratio wanted, and the constant $\langle L\rangle_\Omega$ has cancelled.

4.2 What happens with two scalars instead

Now suppose the map is never applied, and the patch means are taken from the uncorrected signal and divided by the patch-average gains

$$\bar{g}_i \;=\; \langle G\rangle_{R_i}\,, \tag{14}$$

which are two numbers. The estimate becomes

$$r_{\text{scalar}} \;=\; \frac{\langle S\rangle_{R_{\text{sample}}}\,/\,\bar{g}_{\text{sample}}}{\langle S\rangle_{R_{\text{white}}}\,/\,\bar{g}_{\text{white}}}. \tag{15}$$

The only step that can go wrong is $\langle AL\rangle_{R_i} \neq \langle A\rangle_{R_i}\,\langle L\rangle_{R_i}$. The mean of a product is the product of the means plus their covariance, so the relative error of that step is

$$\frac{\langle AL\rangle_{R_i}}{\langle A\rangle_{R_i}\,\langle L\rangle_{R_i}} - 1 \;=\; \rho_i\; c_i^A\; c_i^L, \tag{16}$$

where c_i^A and c_i^L are the coefficients of variation[4] of reflectance and of illumination *within* patch i, and ρ_i is their correlation coefficient, between -1 and $+1$.

So the two-scalar form is exact whenever the target reflectance is uniform within each patch, because then $c_i^A = 0$. Where reflectance is not uniform, the error is the product of three factors, each of which is small.

4.3 How large is that error here?

c_i^L can be measured directly: evaluate the fitted gain map over each patch rectangle and take its coefficient of variation. Median over the 39 white-suite frames, `Gr` layer, $(3, 1)$ fit:

Patch	c_i^L (% of patch mean)	peak-to-peak (%)	$\bar{g}_i$ range over LEDs
`sample`	0.31	1.33	1.066 … 1.139
`chip1`	0.82	3.43	1.042 … 1.104
`white`	1.48	5.75	0.957 … 0.985
`chip2`	2.47	9.02	0.706 … 0.854

[4] **Coefficient of variation.** Standard deviation divided by mean, a dimensionless measure of relative spread. Written as a percentage here.

Inside the sample patch the gain varies by only 0.31 %. Take a pessimistic case: a test strip whose color development varies by 20 % across the patch ($c^A = 20\,\%$), correlated perfectly with the illumination gradient ($\rho = 1$). Eq. 16 then gives $1.0 \times 0.20 \times 0.0031 = 0.062\,\%$, which is below the 0.12 % repeatability floor of §8.1. For the white reference patch, where $c^L = 1.48\,\%$ but the target is a reference surface and c^A should be near 1 %, the same formula gives about 0.015 %.

Two useful observations fall out of that table. The sample patch happens to sit where the illumination is flattest, so the approximation is safest exactly where it matters most. And `chip2`, at the bottom of `bound`, has eight times the sample patch's within-patch gain variation and the largest spread of $\bar{g}$ across LED colors — one more reason it is the worst patch in every test reported later.

4.4 What still needs the map

The two-scalar form is adequate for the concentration measurement itself, but it does not replace the map everywhere.

- **Criterion S and the Layer 3 uniformity flag.** Both are statements about variation *within* a patch, which a patch-average scalar has by definition discarded.
- **Masked or robust patch statistics.** If some pixels in a patch are excluded — hot pixels, a bubble, a defect — the correct scalar is the gain averaged over the surviving pixels, not over the whole rectangle. Computing that requires the map. The same applies to a median or trimmed mean, for which the cancellation argument above does not hold, because those estimators are not linear.
- **Any diagnostic that looks at the corrected image.** The residual heatmaps and profiles of §12 are examples.

Note also what the two-scalar form does *not* avoid: the polynomial itself. $\bar{g}_i$ is the average of the fitted surface over patch i, and there is no way to obtain it without fitting a surface. What the argument removes is the per-pixel division at measurement time, not the calibration.

4.5 Has this been established?

Partly, and it is worth being exact about which part.

Established. Criterion P, Eq. 4, is defined on patch means. It therefore already measures the accuracy of precisely the scalar ratio that Eq. 15 would use. Every number reported in §8 is a statement about that ratio, not about pixelwise flatness. The criterion was designed that way before this question was asked, and the question confirms the design rather than changing it.

Established. The within-patch gain variation c_i^L is measured, and it is small for the sample patch.

Not established. The within-patch reflectance variation c^A of a developed test strip, and its correlation ρ with the illumination gradient, have not been measured. A uniform card cannot measure them, because a uniform card has $c^A \approx 0$ by assumption. That measurement requires imaging developed strips, which is a separate exercise. The bound of 0.062 % above is an estimate under an assumed $c^A = 20\,\%$, not a measurement.

Consequence for the design. Store, per (LED color, Bayer layer, patch role), the single scalar $\bar{g}_i$; that is what the instrument needs at measurement time, and it is a few dozen numbers rather than a few dozen arrays. Keep the polynomial coefficients as the source those scalars are computed from, so that a moved ROI is a recomputation rather than a recapture. Keep the map generation available for diagnostics and for criterion S.

4.6 Why the ratio form explains the multi-field failure

Eq. 15 depends on the two gains only through their *ratio* $\bar{g}_{\text{sample}}/\bar{g}_{\text{white}}$. Any factor common to both cancels. That is why the choice of $\langle F\rangle_\Omega$ in Eq. 11 does not affect the answer when a single field corrects both patches: it is the same factor top and bottom.

It also predicts the failure reported in §8.4. When the sample patch and the white patch are corrected by *different* fields, fitted over different extents, the two normalizing constants $\langle F\rangle_{\Omega_{\text{sample}}}$ and $\langle F\rangle_{\Omega_{\text{white}}}$ are different numbers and do not cancel. Their ratio is exactly the relative brightness of the two extents, which is the quantity being measured. A multi-field scheme therefore cannot produce a patch ratio at all without a further step that ties the fields to a common scale.

5 The model

5.1 Polynomial form

For one combination of (LED color, exposure, Bayer layer), the fitted surface is

$$F(x,y) \;=\; \sum_{i_y=0}^{d_y} \sum_{i_x=0}^{d_x} c_{i_y i_x}\, x^{i_x}\, y^{i_y}, \qquad \text{subject to } i_x + i_y \le \max(d_y, d_x), \tag{17}$$

where d_y and d_x are the maximum exponents allowed on the two axes, $c_{i_y i_x}$ are the coefficients found by least squares, and x and y are pixel coordinates rescaled linearly to the interval $[-1, 1]$.

The rescaling is always taken over the whole `bound` crop, whatever extent the fit was taken over. That keeps coefficient vectors from different extents directly comparable. Fitting is ordinary least squares over the unmasked pixels of Ω.

The constraint $i_x + i_y \le \max(d_y, d_x)$, applied on top of the separate limits d_y and d_x, is what makes the basis *anisotropic*.[5] With $(d_y, d_x) = (3, 1)$ it admits 7 terms,

$$1, \quad x, \quad y, \quad xy, \quad y^2, \quad xy^2, \quad y^3,$$

against 10 for a full bivariate cubic and 15 for a full quartic.

5.2 Why $(3, 1)$

The surface is close to one-dimensional. Averaging each row of the crop and comparing top to bottom gives a fall of about 25 % (denominator: the crop mean). Averaging each column and comparing left to right gives a variation of about 2 % (same denominator). The y axis carries more than ten times the variation of the x axis.

The measured consequence, on the white suite: `P med` improves from 0.24 % at $(3, 1)$ to 0.21 % at $(3, 3)$. That is a gain of 0.03 % of the corrected white-patch mean, against a tolerance of 0.5 %, in exchange for three more free parameters. See Table 1.

5.3 One polynomial per LED color and per Bayer layer

Each LED *color* is a symmetrical array of four emitters. Arrays of different colors cannot occupy the same physical positions, so each color lights the stage from a slightly different geometry, in

[5] **Anisotropic basis.** A basis that allows different maximum powers on the two axes. A conventional bivariate polynomial of degree n is isotropic: it treats x and y alike and costs $\binom{n+2}{2}$ terms. Here the surface varies strongly in y and barely in x, so high powers of x add parameters without adding descriptive power.

addition to part-to-part differences between individual emitters. A separate polynomial per color is therefore expected by construction.

The data agrees. Comparing two normalized fields as 34×34 block means of their ratio (denominator: the frame mean), field shapes differ between LED colors, on the same card, by 0.55 to 2.1 %. On the same measure, `R` and `Gr` field shapes on the *same frame* differ by up to 0.96 %, so layers must be separated as well.

5.4 Masking

Three classes of pixel are excluded before fitting:

- **Saturated**: raw value ≥ 1000 DN. The threshold is $1023 - 23$. The top of the range is already in the sensor's non-linear region, so those counts understate the light that produced them.
- **Non-positive after dark subtraction**: no signal to fit.
- **Hot pixels**, as flagged by `devdesk/roi_darkdiff.py`.

A frame with more than 1 % of its pixels masked (denominator: pixels in the fit extent) is refused rather than fitted, because the surviving pixels are systematically the dimmer ones and a fit over them is biased.

Masking changes conclusions here; it is not routine housekeeping. The stored exposure ladders were metered against a *sample* at the center of the frame, so a white card filling the whole stage oversaturates. In the white suite, `royal_blue/B` has a mean of 959 DN and, without masking, fits with a residual of 0.00 % — a perfect fit to a clipped plateau. `green/e2` in `Gr` and `Gb` is 96 % masked (denominator: pixels in the crop) and is correctly refused; see the `SKIPPED` rows in §12.

6 Criteria for adequacy

The flat field's only job is to make patch *means* comparable between locations. Two consequences shape the criteria, and §4 has already made both precise:

- An error that is constant over a patch cancels in the sample-to-white ratio, because it appears in both numerator and denominator.
- An error that is a linear ramp across a patch does not bias the patch mean, because a plane's departures from its center value are antisymmetric about that center and cancel in the average.

So pixelwise flatness is the wrong basis for a criterion. Judged pixelwise, models that are entirely adequate would fail, and overfitted models would score well. The criteria are defined on patch means instead.

6.1 P — patch-ratio accuracy (primary)

Defined by Eq. 4 in §2.3.2. Require $P \leq \mathrm{tol}_P$, with two tolerances, because they answer different questions:

Mode	tol_P	Rationale
in-sample	0.5 %	about 4× the 0.12 % repeatability floor
transfer	1.0 %	what a stored calibration must achieve in service

Both are percentages of the corrected white-reference patch mean $\bar{s}_{\mathrm{white}}$, which is the same denominator the instrument's own answer uses. P is the criterion that decides acceptance.

6.2 S — residual tilt inside a patch (secondary)

Defined by Eq. 7 in §2.3.3, with the noise reference of Eq. 8. Require $S_i \leq 1\,\%$ of patch i's own corrected mean.

As established in §4, a tilt does not bias the patch mean, so S does not affect the concentration measurement. It matters because Layer 3 of `colorimeter_plan.tex` uses within-patch spread as a strip-uniformity flag, and a leftover illumination tilt would inflate that flag into false alarms.

6.3 T — degree selection (tertiary)

Accept the smallest (d_y, d_x) whose *transfer* P stops improving by more than 0.1 %, that figure being the measured repeatability floor of §8.1.

The purpose of T is to prevent one specific mistake. In-sample residual falls monotonically as degree rises, without limit, because a more flexible model can always follow one capture more closely. Transfer error stops falling once the added terms are describing that capture's own noise and texture rather than the illumination. Selecting on in-sample residual therefore guarantees an overfitted model. The harness prints in-sample and transfer side by side in every table so the difference between them stays visible.

7 The extent ladder

"Fit over an area larger than the target" leaves open how much larger. Five rungs were defined, cheapest first, to be climbed only as far as necessary:

1. `bound` — the whole crop, 330×98 layer pixels.
2. `bound_trim05` and `bound_trim10` — the crop less a 5 % or 10 % border on each side (denominator: the crop's own height and width).
3. Whole frame — from the older `*_img.h5` captures.
4. `local_<patch>` — the patch grown by 50 % on every side, about 51×51 layer pixels, fitted at low degree.
5. `sector<k>_<i>` — the crop split into k horizontal bands, with no continuity enforced between bands. The split is along y because that is the axis the illumination varies on.

Rungs 4 and 5 turn out to be unusable for criterion P, for the structural reason given in §4.6 and reported in §8.4.

Rung 3 remains untested. The capture suites store only the `bound` crop. Whole-frame raw data exists only in the older `*_img.h5` files, which carry neither a dark frame nor an LED label. Testing rung 3 properly needs a configuration switch on the normal capture path to archive full-image Bayer layers *and* full-image dark Bayer layers. That is a change in `src/` and has deliberately not been made.

8 Results

Two capture suites, both of plain colored card filling the stage, taken 2026-08-03:

Suite	Stage	Frames scored
`white_20260803_1742.h5`	white card stock	39
`mgray_20260803_1736.h5`	gray card stock	40

One frame is scored per (LED color × Bayer layer): the brightest exposure in the ladder that is not clipped. §8.8 discusses that selection rule and proposes a tightening of it.

8.1 Repeatability floor

Before any residual can be interpreted, the size of a measurement that means "zero" has to be known. Two comparisons that ought to be flat were measured, as the 34×34 block RMS of a ratio between two normalized frames (denominator: the frame mean), over frames brighter than 300 DN:

Comparison	white suite	gray suite
`Gb` against `Gr`, same frame	0.118 %	0.101 %
Adjacent exposures, same LED	0.128 %	0.173 %

Shot noise alone on a 1156-pixel mean predicts about 0.04 %, so these figures are above the pure counting limit but they are the practical floor of the whole pipeline. Take 0.12 % as that floor. Nothing at or below it is a measurement. The 0.5 % in-sample tolerance sits about 4× above it.

8.2 Magnitude of the uncorrected non-uniformity

Across the crop, before correction, the illumination spans about +7 % to −17 % of the crop mean. The `chip2` patch, at the bottom of `bound`, sits in the worst of it. That span is the size of the error the flat field exists to remove.

8.3 In-sample ladder

Table 1: White card suite, 39 frames. Columns as defined in §2.3: `P med` and `P max` are the median and maximum, over frames, of each frame's worst-patch P; `S med` is the median worst-patch S; `S/nz` is that tilt divided by its noise reference; `blk rms` is the block residual. Terms is the number of free coefficients.

Extent	(d_y, d_x)	Terms	P med	P max	S med	S/nz	blk rms	P
`bound`	(1,1)	3	8.86	16.35	5.67	10.62	3.77	fail
`bound`	(2,1)	5	0.88	1.59	1.68	4.13	0.47	fail
`bound`	(2,2)	6	0.90	1.64	1.72	4.09	0.41	fail
`bound`	(3,1)	7	**0.24**	**0.46**	1.68	4.32	0.38	**PASS**
`bound`	(3,2)	9	0.21	0.41	1.74	4.49	0.28	PASS
`bound`	(3,3)	10	0.21	0.41	1.76	4.60	0.28	PASS
`bound`	(4,2)	12	0.28	0.50	1.48	3.68	0.19	PASS
`bound`	(6,2)	18	0.28	0.58	1.21	2.80	0.15	fail
`bound_trim05`	(2,1)	5	1.11	2.50	1.91	4.73	0.45	fail
`bound_trim05`	(3,1)	7	0.39	1.04	1.79	4.51	0.39	fail
`bound_trim10`	(3,1)	7	0.93	1.97	2.15	5.57	0.49	fail
`sector3`	(1,1)	3	—	—	1.22	2.85	—	n/a
`sector3`	(2,1)	5	—	—	1.08	2.96	—	n/a
`local`	(1,1)	3	—	—	1.11	2.59	—	n/a
`local`	(2,1)	5	—	—	1.19	3.10	—	n/a
`local`	(2,2)	6	—	—	1.18	3.09	—	n/a

Four results follow from these tables.

Degree 1 is inadequate; degree 3 is where the improvement stops. A plane leaves 8.9 to 9.6 % median P, so the surface genuinely curves. Raising d_y from 2 to 3 gains 0.64 % on the white card. Raising it from 3 to 6 gains nothing.

Table 2: Gray card suite, 40 frames. Same columns. No rung passes criterion P even in-sample; see §8.7.5.

Extent	(d_y, d_x)	Terms	P med	P max	S med	S/nz	blk rms	P
`bound`	(1,1)	3	9.63	16.63	6.99	6.11	4.18	fail
`bound`	(2,1)	5	1.20	7.51	3.30	2.57	0.55	fail
`bound`	(2,2)	6	1.12	7.47	3.30	2.57	0.49	fail
`bound`	(3,1)	7	0.73	1.29	2.58	2.02	0.49	fail
`bound`	(3,2)	9	0.72	1.26	2.52	1.98	0.42	fail
`bound`	(3,3)	10	0.72	1.26	2.73	2.12	0.42	fail
`bound`	(4,2)	12	0.55	1.08	2.40	2.03	0.37	fail
`bound`	(6,2)	18	0.46	1.00	2.47	2.07	0.33	fail
`bound_trim05`	(3,1)	7	1.08	2.23	3.10	2.36	0.50	fail
`bound_trim10`	(3,1)	7	1.09	1.85	2.88	2.10	0.67	fail

Raising d_x changes nothing. $(3,1)$, $(3,2)$ and $(3,3)$ differ by at most 0.03 % in median P. This is the direct evidence for the anisotropic basis.

Overfitting is visible in-sample. On the white card, P at $(6,2)$ is 0.28 % median and 0.58 % maximum, which is worse than $(3,1)$'s 0.24 and 0.46, while the block residual continues to fall from 0.38 to 0.15 %. The added terms are describing the card's own texture, and they do so near the patches, where it costs most. A criterion based on pixelwise flatness would have selected $(6,2)$.

Trimming the fit extent makes the result worse. `bound_trim05` at $(3,1)$ scores 0.39 % against `bound`'s 0.24 %, and `trim10` scores 0.93 %. The patches sit near the edges of the crop, so trimming forces the polynomial to extrapolate exactly where it is being judged. Rung 2 of the ladder is rejected.

8.4 Local and sector fits cannot produce a patch ratio

Rungs 4 and 5 report no P at all. This is the structural consequence derived in §4.6: each local or sector field is normalized over its own extent, so the two normalizing constants do not cancel in the ratio, and their quotient is exactly the relative brightness of the two extents — which is the quantity being measured.

The harness reports these entries as blank rather than computing a number that would look meaningful and be arbitrary. Their S column is still informative, and local fits do flatten a patch well, but S is the secondary criterion. Any scheme using more than one field needs a step that ties the fields to a common scale before it can be used at all.

8.5 Transfer

Fields fitted on the white suite, applied to the gray suite. Both P columns use the corrected white-patch mean as denominator. The `aligned` columns repeat the transfer with the field displaced bodily by the whole number of layer rows that suits it best; §8.6 explains why that column exists.

The difference between the two halves of the table is the point. In-sample P improves by a factor of four from $(1,1)$ to $(3,1)$ and then stops. Transfer P is best at $(2,1)$ and grows worse as degree rises. Criterion T, applied mechanically, selects $(2,1)$.

That selection should not be acted on, for the reason given in §8.7.4: the two suites imaged

Table 3: Transfer: white → gray, 33 (LED × layer) pairs after exclusions. Violet excluded, per §8.7.1.

		in-sample		transfer		aligned		
(d_y, d_x)	Terms	P med	blk	P med	P max	P med	P max	@row
(1,1)	3	8.98	3.81	10.14	16.51	10.06	16.24	+14
(2,1)	5	0.92	0.46	**0.98**	1.90	**0.79**	1.66	+2
(3,1)	7	0.23	0.38	1.34	2.05	1.13	1.95	+2
(4,1)	9	0.27	0.32	1.41	2.15	1.24	2.01	+2
(4,2)	12	0.27	0.19	1.40	2.14	1.24	2.00	+2
(5,2)	15	0.26	0.16	1.32	2.07	1.13	1.90	+2
(6,2)	18	0.28	0.15	1.38	2.20	1.20	2.03	+2

different physical stages. A transfer between captures of two different cards measures card differences and model error at the same time and cannot separate them. Table 3 should be read as an upper bound — transfer error is no worse than about 1 % median — with the ranking of degrees within it not yet trustworthy.

8.6 Registration offset

The shift scan displaces the fitted field by a whole number of layer rows and rescores. At $(2, 1)$:

Offset (layer rows)	−8	−4	−2	0	+2	+4	+8
Median transfer P (%)	2.14	1.58	1.28	0.98	**0.79**	0.84	1.59

There is a clear minimum at +2 layer rows, which is +4 stream pixels, improving median P from 0.98 to 0.79 %. That is a registration offset between the two captures — the card or the stage moved — and no polynomial degree can correct it. Only re-registering the captures can.

The scan exists because without it the degree comparison is confounded. A low-degree field can score well on a transfer test simply by being too smooth to respond to a misalignment. Comparing degrees at each one's own best alignment removes that advantage.

8.7 Confounders found and handled

8.7.1 Optical brighteners in the white card

Under violet illumination the white card returned 22.4× the signal of the gray card, where every other LED color agreed on a ratio near 3.7×. (Denominator in both figures: the gray card's dark-subtracted patch mean.) A sixfold departure from the consensus ratio is not a reflectance difference.

White card stock contains optical brighteners: fluorescent dyes that absorb near-ultraviolet light and re-emit it in the blue. Under violet LEDs the card is a light *source*, and the surface fitted to it is the emission pattern of the paper, which will not transfer to any other target.

The harness detects this automatically, in `find_anomalous_leds()`, using a threshold of 2× the consensus ratio. It reports the finding and excludes the offending LED from the pooled figures. That is why Table 3 states "violet excluded" rather than silently carrying a 20 % outlier.

8.7.2 Green crosstalk under red LEDs

`Gb` and `Gr` agree to 0.10–0.12 % under most LEDs, which is the repeatability floor. Under `photo_red`, `red` and `red_orange` they disagree by more than 1 % on the same block-RMS

measure (denominator: the frame mean), reproducibly at every exposure, so the disagreement is not noise.

The green photosites that neighbor red pick up different optical and electrical crosstalk from those that neighbor blue. The consequence is firm: fit each Bayer layer separately, and never average the two greens. Agreement between `Gb` and `Gr` remains a useful internal control, but only for the non-red LEDs.

8.7.3 Card texture, and how much of the residual it explains

The residuals reported above are limited partly by the card, not only by the model, and the two can be separated.

Take two exposures of the same LED and layer, scale each to its own mean, and form their ratio. Structure that is fixed on the card appears identically in both frames and cancels in that ratio; random noise does not cancel, and adds in quadrature, so the ratio's spread divided by $\sqrt{2}$ estimates the noise alone. Subtracting that from a single frame's spread, in quadrature, leaves the fixed structure.

At the pixel scale, on the sample patch, with both cards at comparable brightness:

Suite / LED / layer	mean DN	pixel CV	noise part	fixed part
white / `white` / `B`	526	1.44 %	1.12 %	0.89 %
white / `cyan` / `B`	632	1.34 %	1.09 %	0.78 %
gray / `royal_blue` / `B`	532	4.01 %	1.29 %	3.80 %
gray / `violet` / `B`	580	3.51 %	1.15 %	3.32 %

(All percentages are of the patch mean.) At the same signal level the random noise is the same on both cards, near 1.1 %, as it must be if it is shot noise. The fixed part is four times larger on the gray card. The gray card stock is substantially more textured than the white.

At the 34×34 patch scale, after removing the fitted field, the same separation gives:

Suite / LED / layer	block CV	noise part	fixed part
white / `white` / `B`	0.394 %	0.091 %	0.383 %
white / `green` / `Gr`	0.375 %	0.084 %	0.365 %
white / `blue` / `Gr`	0.353 %	0.045 %	0.350 %
gray / `royal_blue` / `B`	0.328 %	0.145 %	0.294 %

(All percentages are of the frame mean, per Eq. 3.) This is the more important of the two tables. The block residual after a $(3, 1)$ fit — 0.38 % for the white suite in Table 1 — is almost entirely fixed structure on the card, with only 0.05 to 0.09 % attributable to noise.

Two conclusions follow. The polynomial is performing better than the block-residual figure suggests, because most of that figure is the target rather than the model. And criterion P, which assumes the imaged card is uniform, is measuring card structure and fit error together whenever the card has structure of its own. Separating them is exactly what the `grayR180` capture of §11 is for.

8.7.4 Staging differences between the two suites

The white and gray suites are not two captures of one scene. They used different sheets of card stock, separately placed. Paper texture, thickness, flatness and placement all differ at once, and the +2-row registration offset confirms that placement did.

There is a second staging issue. The production chamber has a flat-black stage and curtains except at the sample, whereas a card filling the whole stage scatters substantially more light back into the optical path than the instrument will normally see. The surfaces fitted here therefore include stray light that the instrument does not have.

Neither issue invalidates the analysis. The in-sample results, the model form, the criteria and the confounders are all sound. Both are reasons the transfer figure is an upper bound. Measuring the true transfer error requires the controlled protocol of §11.

8.7.5 The gray suite fails P in-sample

Table 2 shows no rung passing criterion P on the gray card: best case 0.46 % median but 1.00 % maximum at $(6, 2)$, and 0.73/1.29 % at $(3, 1)$. The white card passes comfortably at the same degree.

Signal level is the likeliest explanation. The gray card returns 0.27× the white card's signal, so its dim layers sit where noise is a much larger fraction of the patch mean. §8.8 tabulates this, and the worst transfer pairs in §8.5 are readings at 7 to 14 DN on the gray card.

Card structure is the other candidate, but §8.7.3 weighs against it: at patch scale the gray card's fixed structure (0.29 %) is no worse than the white card's (0.35 to 0.38 %), even though at pixel scale it is four times larger. The texture appears to be fine-grained enough to average out over a 34×34 patch.

The `grayA`/`grayA2` pair of §11 resolves this, and the case for re-metering the exposures against the calibration card is strong either way.

8.7.6 Enlarging `bound` makes the result worse

This was tested directly, because enlarging `bound` had been proposed as a way to give the polynomial more support.

At fixed degree, patch error grows as the fit area grows. On the gray card the worst-patch P went from 0.46 % over a 330-row crop to 2.61 % over an enlarged one; on a light-gray capture, from 0.17 % to 3.06 %. Raising the degree did not recover it. The result was confirmed independently on a later 530-row capture: fitting over 530 rows costs 1.3 to 2.1× more residual within the same central region than fitting over 330 rows does.

A polynomial distributes its degrees of freedom over the whole fit extent. Enlarging the extent asks the same 7 terms to describe more of a curve, and the added territory is territory nothing is measured in. The recommendation is the opposite one: move the reference patches inward, toward the sample, rather than moving `bound` outward. `chip2` sits at the bottom of `bound`, where illumination has fallen 17 %, and it is the worst patch in every transfer test.

8.8 Exposure ladder selection

`capture_runner` records several exposures per LED in a suite, and some are expected to saturate somewhere in the frame. Discarding those and requiring only that one usable exposure survive is the right policy, with one refinement.

The selection must be per (LED, layer), not per LED. The harness already does this, in `select_frames()`: for each (LED color, Bayer layer) independently it takes the unclipped exposure with the highest mean. Selecting per LED would be wrong, because the four layers of one frame differ enormously in level. Under the green LED at exposure `e2`, `Gr` and `Gb` are 96 % masked while `R` sits at 66 DN and `B` at 220 DN, both perfectly usable. Discarding the whole frame would throw away good data in two layers out of four.

"Unclipped" is necessary but not sufficient. An unclipped frame can still be too dim to be worth fitting. Averaging n pixels reduces relative noise by $\sqrt{n}$, so the relative noise on a patch mean is

$$\varepsilon_i = \frac{\sigma_i}{\sqrt{n_i}}, \tag{18}$$

where σ_i is the pixel coefficient of variation within the patch, in percent, and $n_i = 1156$. Measured on the sample patch of the selected frames:

Suite	patch mean, DN	pixel CV σ	patch-mean noise ε
white	27.5	5.87 %	0.173 %
white	70.9	3.20 %	0.094 %
white	241.3	2.00 %	0.059 %
white	903.5	1.12 %	0.033 %
gray	7.8	15.59 %	0.459 %
gray	45.2	5.66 %	0.167 %
gray	310.8	4.46 %	0.131 %
gray	579.8	3.51 %	0.103 %

(All percentages are of the sample patch's own mean.) The gray card never reaches $\varepsilon = 0.1\,\%$ at any exposure in the suite, while the white card reaches it above about 70 DN. That is a concrete, quantitative statement of why the gray suite underperforms, and it is a direct argument for re-metering the exposure ladder against the calibration card.

Proposed success criterion. A suite is adequate for flat-field work when, for every (LED color, Bayer layer) combination the analysis consumes, at least one exposure in the ladder satisfies both of:

1. masked fraction $\leq 1\,\%$ of the fit extent (already enforced, §5.4);
2. $\varepsilon \leq 0.1\,\%$ by Eq. 18, which at $n = 1156$ means a pixel CV of $\sigma \leq 3.4\,\%$ (not yet enforced).

Exposures that fail either test are discarded, not repaired, and there is no requirement that one exposure serve all four layers. The second test is a small addition to `select_frames()` and would turn the present silent underperformance into an explicit report.

Which combinations must be consumed. The five worst transfer pairs of §8.5 — `amber/B`, `blue/R`, `photo_red/B`, `royal_blue/R`, `red_orange/B` — are all combinations in which the LED color and the Bayer layer do not match, and all of them read between 7 and 14 DN on the destination card. They are crosstalk readings carrying almost no signal. If the measurement consumes only the matching combinations, the success criterion need only apply to those, and the pooled transfer figures should improve substantially once the mismatched ones are excluded. Testing that requires a layer filter on the harness, which it does not yet have.

9 Choices made

Question	Choice	Basis
Correction form	divide by unit-mean gain	§3.1
Basis	anisotropic (d_y, d_x)	row vs column variation, 25 vs 2 %
Default degree	$(3, 1)$, 7 terms	Table 1
Fit extent	the whole crop	trims and enlargements both worse
Granularity	per LED color, per layer	2.1 % and 0.96 % shape differences
Normalization	mean over the fit extent	§4.6
Coordinates	$[-1, 1]$ over `bound` always	comparable coefficients
Degree selection	aligned transfer P	criterion T
Runtime form	per-patch scalar gains	§4
Frame selection	best unclipped, per (LED, layer)	§8.8
Calibration card	brightener-free and smooth	§8.7.1, §8.7.3
Storage	coefficients authoritative	maps and scalars derived

$(3, 1)$ is the default, not a settled answer. Criterion T on the present data prefers $(2, 1)$, but on data that cannot separate model error from card differences. The choice between $(2, 1)$ and $(3, 1)$ is to be re-run on the controlled suites.

10 Implementation

10.1 `devdesk/flatfield.py`

A desktop-only evaluation harness, sibling of `hdf5inspect.py` and `roi_darkdiff.py` in the same directory, sharing their conventions: argparse mode flags, `Hdf5Reader` from `src/hdf5read.py`, bare filenames resolved against `captures/`, results into `devdesk/devdesk.sqlite`. It makes no changes to `src/` behavior.

Principal types:

`Frame`
: One dark-subtracted, masked layer, with its LED, exposure label, exposure in microseconds and dark link. `.usable` is false when more than 1 % of pixels are masked.

`Patches`
: The active patch rectangles in layer coordinates, plus which one holds the white-reference role. Loaded through `roi.load_active()` and `roi.to_layer_rect_within()`; never hardcoded.

`FlatField`
: Coefficients, degrees, normalization domain, fit extent, the window it covers, and `norm_mean` $= \langle F \rangle_\Omega$. Provides `gain()`, `correct()`, `gain_for_patch()` and `coef_json()`.

`Model`
: A named collection of one or more `FlatField`s with `field_for(rect)`, so that single-field and multi-field extents are scored through one interface.

`Score`
: P, S, S/S^{nz}, block residual, and the count of patches no field could cover without extrapolating.

The `covers` field on `FlatField` deserves a note, because getting it wrong produced a wrong answer once. `bound` extents cover the whole crop; `local_*` and `sector*` extents cover only their own window. Without that distinction, trimmed fits were scored on whichever patches happened to fall inside the trim window, which is a smaller and easier subset, and trimming looked better than it is.

10.2 Storage

Coefficients are the source of truth; maps and scalars are regenerable caches.

Table `flatfield` in `devdesk/devdesk.sqlite`, keyed on (`h5file`, `led`, `eno`, `bayerlayer`, `extent`, `deg_y`, `deg_x`), holds `coef_json`, the normalization domain (`domain_rows`, `domain_cols`, `bound_*`, `frame_*`), `norm_mean`, and the fit statistics (`n_fit`, `n_masked`, `fit_rms_pct`, `block_rms_pct`, `patch_dev_max_pct`).

Derived gain maps go to

```
devdesk/flatfield/<suite_id>/<led>_<ch>_<roi>_gain.npy
```

holding the field evaluated over one patch rectangle, divided by its own mean. This is the artifact `colorimeter_methods.tex` called `color_bayer_rect_trend.npy`, changed from additive to multiplicative, and regenerable from the coefficients whenever a patch moves. That regenerability is the reason for keeping the coefficients authoritative.

§4 adds a third, smaller artifact for the runtime path: the per-patch scalar $\bar{g}_i$, which is the mean of that same gain map.

10.3 Archive metadata (capture schema 2)

The analysis needs to know the `bound` rectangle a capture was taken with. The stored layer shape gives its width and height; it cannot give its x and y, and a uniform card carries no features to register against.

Consulting the local `roi` table is not a substitute. The analysis machine's `colorimeter.sqlite` is a copy, and an ROI edit made on the instrument never reaches it. This produced a real false alarm: an apparent fault in the HDF5 writer that was in fact a stale desktop database.

Archives now describe themselves. `capture_h5.SCHEMA_VERSION` is 2, and each archive carries:

- root attributes `bound_x`, `bound_y`, `bound_w`, `bound_h`, which are cheap to read and sufficient to interpret the layers;
- a `/meta` JSON record with the full patch set, camera settings, and rows from the `device` and `samples` tables.

`src/capture_meta.py` collects that record declaratively. Column lists and field extractors are tuples at the top of the module, so adding a field is a one-line edit in one place. `collect()` never raises; a block that fails becomes `{"error": ...}`, because a metadata problem must never cost a capture.

`flatfield.py` reads `archive_bound()` first and falls back to the `roi` table only for schema-1 archives, printing a warning when it does. A `--bound X,Y,W,H` override handles archives taken against geometry this database does not have.

Deployment note. The schema-2 writer must be deployed to the Pi before the protocol captures of §11 are taken. Archives written by the current Pi code are still schema 1 and will not carry their own geometry.

11 Image sampling protocol

The binding uncertainty is the transfer error, and it cannot be resolved from the existing captures, because those used two different stages. Four short suites on one card resolve it.

11.1 Card selection

The card must satisfy two independent requirements, and the earlier recommendation to use gray stock addressed only the first.

1. **Free of optical brighteners.** White stock fluoresces under violet (§8.7.1), and its surface under that LED is the paper's emission rather than the instrument's illumination. Check by confirming that the violet LED's brightness ratio against a reference card sits with the other LEDs rather than several times above them.
2. **Smooth at the pixel scale.** The gray stock measured in §8.7.3 carries 3.3 to 3.8 % fixed texture at pixel scale, four times the white stock's. Check by the two-exposure ratio method of §8.7.3, which separates fixed texture from random noise; aim for a fixed part below 1.5 % of the patch mean.

The gray stock's texture largely averages out over a 34×34 patch, so it is usable, but a card meeting both requirements gives a cleaner measurement of the instrument.

11.2 Preparation

1. Select and verify the card as above.
2. **Mark one corner** of the card, indelibly and visibly in frame if possible. The `grayR180` suite depends on knowing the orientation.
3. **Re-meter the exposure ladder against this card**, so that the maximum DN anywhere inside `bound` is ≤ 850 DN in every layer. Metering on a center patch is what oversaturated the earlier white suite. A ceiling of 850 leaves margin below the 1000 DN mask threshold for the brighter corners.
4. Confirm that the ladder meets the success criterion of §8.8: for every (LED, layer) combination consumed, at least one exposure with $\sigma \leq 3.4$ % on the sample patch.
5. **Deploy schema-2 capture code to the Pi** first, so that each archive records its own `bound`.
6. Let the LEDs and the enclosure reach thermal steady state before the first suite, and take all four suites in one sitting.

11.3 The four suites

Suite	Action before capture	What it isolates
`grayA`	place the card, capture	baseline
`grayA2`	**touch nothing**, capture again	noise floor alone
`grayB`	lift the card, replace it carefully	placement repeatability
`grayR180`	rotate the same card 180° in plane	paper against instrument

Each suite is the full LED set with the re-metered ladder. It is the same capture the instrument already performs, with a card on the stage.

11.4 What each suite answers

`grayA` → `grayA2` should transfer at the repeatability floor, about 0.12 % of the white patch mean. If it does not, something in the pipeline is contributing error of its own and every other figure here is suspect. This is the control.

`grayA` → `grayB` adds placement. The difference between this result and `grayA2`'s is the cost of re-seating a card, and it sets a floor on how well any calibration can be verified by re-imaging a target.

`grayA` → `grayR180` is the decisive one. Rotating the card 180° leaves the instrument's illumination pattern untouched but rotates the paper's own structure with the card. Therefore:

- Structure that stays in place under rotation belongs to the instrument. It is what the polynomial is for, and it is correctable.
- Structure that rotates with the card belongs to the paper. No flat field can remove it, and it is a hard floor on how well criterion P can ever be verified, because P assumes the card is uniform and this measures the extent to which it is not.

That decomposition converts Table 3's upper bound into an actual measurement of model error, and it settles the choice between $(2, 1)$ and $(3, 1)$. §8.7.3 has already shown the effect is large enough at the pixel scale to be worth separating, and this is the measurement that separates it at the patch scale.

11.5 Optional additions

If a configuration switch for full-image Bayer layers plus full-image dark layers is added to the normal capture path, take one extra `grayA` suite with it enabled. That makes ladder rung 3 testable and answers directly whether `bound` is a large enough support region.

12 Running the programs

Use the project virtual environment. The system Python lacks `h5py`, `cv2` and `matplotlib`.

```
cd /home/bill/Projects/Colorimeter
.venv/bin/python3 devdesk/flatfield.py --selftest
```

Modes:

Flag	Purpose
`-f`, `--fit`	fit the default $(3, 1)$ over `bound`; print coefficients
`-c`, `--compare`	the extent × degree ladder, in-sample
`-t`, `--transfer`	fit on FILE, apply to FILE2
`-p`, `--plot`	profiles, residual heatmap, error-against-degree curves
`-w`, `--write`	store coefficients and derived gain maps

Filters and overrides: `--led COLOR` (repeatable), `--exp LABEL` (repeatable), `--deg-y`, `--deg-x`, `--bound X,Y,W,H`, `--captures-dir`, `--db`, `--gain-dir`, and `-o FILE.png` to save a `-p` figure instead of showing it.

12.1 Fitting one LED

```
.venv/bin/python3 devdesk/flatfield.py white_20260803_1742.h5 -f --led green
```

```
white_20260803_1742.h5: flat-field fits, (deg_y, deg_x) = (3, 1), 7 terms
  basis: y^0x^0 y^0x^1 y^1x^0 y^1x^1 y^2x^0 y^2x^1 y^3x^0
  led      e#  ch   exp_us    mean  masked%   span%  resid%  coef[0..2]
  green    e0  R      5291    29.1    0.00%   29.1%   5.86%    30.47  -0.43   -3.64
  green    e0  Gr     5291   514.1    0.00%   31.2%   1.59%   538.65  -6.52  -69.48
  green    e0  Gb     5291   514.0    0.00%   31.3%   1.59%   538.66  -6.62  -69.87
  green    e0  B      5291    97.3    0.00%   35.8%   3.02%   102.40  -1.51  -16.22
  green    e1  Gr     7937   770.9    0.00%   31.1%   1.45%   808.37  -9.46 -102.98
  green    e2  Gr    11906   918.1   96.24%  SKIPPED - over 1% masked
```

Column meanings:

`mean`
: $\langle S \rangle_{\text{crop}}$, the dark-subtracted mean over the crop, in DN. Use it to judge whether a row is worth reading at all. At 29 DN the `R` layer under a green LED is nearly all noise, which is why its `resid%` is 5.86 while the well-exposed `Gr` row's is 1.59.

`masked%`
: Percent of crop pixels excluded. Denominator: all pixels in the crop. Above 1 % the row is `SKIPPED`.

`span%`
: $(\max F - \min F)$ divided by $\langle F \rangle_{\Omega}$, over the fitted surface. It is the size of the illumination variation the fit describes, about 30 % here.

`resid%`
: Root-mean-square of the pixelwise fit residual, as a percentage of the frame mean. This is a *pixelwise* figure and is dominated by shot noise. Do not compare it with criterion P, which is a patch-mean figure and roughly $\sqrt{1156}$ times smaller for the same underlying error.

`coef[0..2]`
: The first three coefficients: the constant, the x term and the y term. The y coefficient is about $10\times$ the x coefficient with the same sign pattern in every row, which is the anisotropy in raw form.

12.2 The ladder

```
.venv/bin/python3 devdesk/flatfield.py mgray_20260803_1736.h5 -c
```

Prints the repeatability floor, the patch positions in use, then the content of Table 2. Read the floor block first; it states what the smallest meaningful number in the table is.

12.3 Transfer

```
.venv/bin/python3 devdesk/flatfield.py grayA_<stamp>.h5 grayR180_<stamp>.h5 -t
```

Prints the fluorescence exclusion list, the degree table (Table 3), the criterion-T recommendation, the shift scan, and the five worst (LED × layer) pairs with their source and destination brightness in DN.

12.4 Writing results

```
.venv/bin/python3 devdesk/flatfield.py grayA_<stamp>.h5 -w
```

Writes `flatfield` rows and `*_gain.npy` maps, then verifies the round trip by rebuilding a gain map from the stored `coef_json` and comparing it with the file.

13 Interpreting results

13.1 Reading order

1. **The floor block.** If `Gb` against `Gr` is not near 0.12 %, stop. The pipeline is contributing error and nothing below it is trustworthy.
2. **The exclusion warnings.** A fluorescence flag or a `SKIPPED` row changes which frames the pooled figures cover.
3. **The patch list.** Confirm the patch positions are the intended ones; a stale `roi` table moves them without any other symptom.
4. **Transfer P, not in-sample P**, for any decision about degree.
5. **`P max`, not `P med`**, for any decision about acceptance. The median conceals the one patch that fails.

13.2 Decision rules

Transfer P falls with degree, then flattens.
The expected pattern. Take the degree at which it flattens, per criterion T.

Transfer P rises with degree while in-sample P falls.
Overfitting. Take the lower degree. This is what Table 3 shows above $(2, 1)$.

The shift scan has a minimum away from zero.
The two captures are not registered to each other. Correct the staging or the registration; changing the polynomial cannot help. Compare degrees on the `aligned` columns only.

S is large but `S/nz` is near 1.
The tilt is a plane fitted through noise, not a real tilt. Ignore it. This is common in dim layers, where the pixel coefficient of variation reaches 6 %.

S is large and `S/nz` is well above 1.
A real residual tilt that no whole-crop polynomial removed. It does not bias the patch mean, so P can still pass, but it will inflate Layer 3's uniformity flag.

Block residual keeps improving while P stalls or worsens.
The added terms are describing texture, not illumination. Trust P. Compare the block residual against §8.7.3's figures before concluding the model is at fault: most of that residual is the card.

P is blank.
A multi-field model. Not a missing number; see §8.4.

A single LED is a large outlier.
Check fluorescence first, then signal level. The worst-pairs listing prints source and destination DN precisely so that a 1.6 % error at 11 DN can be recognized as noise rather than field error.

13.3 What adequate will look like

On the controlled suites, the flat field is adequate when all of:

- `grayA` → `grayA2` transfer `P max` ≤ 0.3 % of the corrected white patch mean, which is near the floor;
- `grayA` → `grayB` transfer `P max` ≤ 1.0 %;
- in `grayA` → `grayR180`, the part of the residual that stays fixed under rotation is ≤ 1.0 %, the rotating part being card structure rather than model error;
- the shift scan on `grayA` → `grayA2` has its minimum at zero rows;
- the chosen degree is the point at which aligned transfer P flattens, not the point at which in-sample residual flattens.

14 How this fits into next steps

14.1 Settled

- Model form: multiplicative correction, anisotropic polynomial, one per LED color and per Bayer layer, fitted over the whole crop, normalized over its fit extent.
- Criteria defined and implemented: P, S with its noise reference, and T on aligned transfer.
- The correction reduces at measurement time to two scalars per patch pair (§4), with the error of that reduction bounded by a product of within-patch variations.

- Extent ladder mostly resolved: the whole crop wins; trims are worse; multi-field schemes cannot produce a patch ratio without a cross-normalization step; whole-frame is untestable with current archives.
- Confounders identified and handled in code: saturation, optical brighteners, red-LED green crosstalk, registration offset, and card texture.
- Archives made self-describing (capture schema 2), so analysis no longer depends on a database copy staying in step.
- Patch selection converted from hardcoded names to roles, so that renaming a patch cannot silently misdirect the analysis.

14.2 Immediately next

1. Deploy schema-2 capture code to the Pi.
2. Run `src/arbiter.py`'s self-test on the Pi. It cannot run on the desktop, which lacks `gpiozero`, so the role-based edits to it are compile-checked but not executed.
3. Add the $\varepsilon \leq 0.1\,\%$ test to `select_frames()` (§8.8), and a layer filter so that mismatched (LED, layer) combinations can be excluded from pooled figures.
4. Select and verify a calibration card against both requirements of §11.
5. Re-meter the exposure ladder against it to the 850 DN ceiling.
6. Capture `grayA`, `grayA2`, `grayB`, `grayR180`.
7. Re-run `-c` and `-t` across them, and settle $(2, 1)$ against $(3, 1)$ on data that separates model error from card structure.

14.3 Then

- **Promote the evaluated model into `src/`.** The harness stays in `devdesk/` as the evaluation tool. The instrument needs only to look up two stored scalars per patch pair, with the map generation available for diagnostics.
- **Measure within-patch reflectance variation on developed strips**, which is the one unmeasured factor in the bound of §4.
- **Decide the calibration cadence:** how often the field is refitted, and against what — a stored card, or the white reference patch on each strip.
- **Move the reference patches inward.** `chip2` sits where illumination has fallen 17 %, it has eight times the sample patch's within-patch gain variation, and it is the worst patch in every test. Bringing the references nearer the sample reduces the flat field's workload substantially. This is a layout change with consequences beyond this task.
- **Bayesian update within the patch:** a stored field as prior, updated by the few measurements each run actually makes, rather than re-estimated from scratch. Deferred deliberately, because it is worth building only once the prior's quality is known, and that is what the protocol captures establish.
- **Full-frame capture switch**, to close ladder rung 3.

14.4 Relation to the other documents

This is Layer 1 of `colorimeter_plan.tex`, with two amendments to `colorimeter_methods.tex`: the correction divides rather than subtracts (§3.1), and the stored artifact is a multiplicative gain derived from authoritative coefficients rather than an additive trend array (§10). Layer 3's strip-uniformity flag is the consumer of criterion S.